23/sep/2020

1. Find the Fourier Cosine Transform of

$$f(x) = \begin{cases} x & \text{for } 0<x<1 \\ 2-x & \text{for } 1<x<2 \\ 0 & \text{for } x>2 \end{cases}$$

$$F_c\{f(t)\} = \sqrt{\frac{2}{\pi}} \int_0^\infty f(t) \cos st \, dt$$

$$= \sqrt{\frac{2}{\pi}} \left[\int_0^1 t \cos st \, dt + \int_1^2 (2-t) \cos st \, dt \right]$$

$$= \sqrt{\frac{2}{\pi}} \left[\left[t \frac{\sin st}{s} - \frac{\cos st}{s^2} \right]_0^1 + \left[(2-t) \frac{\sin st}{s} + \frac{\cos st}{s^2} \right]_1^2 \right]$$

$$= \sqrt{\frac{2}{\pi}} \left[\left[\frac{\sin s}{s} - \frac{\cos s}{s^2} + \frac{1}{s^2} \right] + \left[\frac{\cos 2s}{s^2} - \frac{\sin s}{s} - \frac{\cos s}{s^2} \right] \right]$$

$$= \sqrt{\frac{2}{\pi}} \left[\cancel{\frac{\sin s}{s}} - \frac{\cos s}{s^2} + \frac{1}{s^2} + \frac{\cos 2s}{s^2} - \cancel{\frac{\sin s}{s}} - \frac{\cos s}{s^2} \right]$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{1 + \cos 2s - 2\cos s}{s^2} \right]$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{\cancel{1} + 2\cos^2 s - \cancel{1} - 2\cos s}{s^2} \right]$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{2\cos s \, (1 - \cos s)}{s^2} \right]$$

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Q. $f(x) = \begin{cases} 1 & \text{for } 0 < x < a \\ 0 & \text{otherwise.} \end{cases}$ Fourier Cosine Transform.

$$F_c\{f(x)\} = \sqrt{\frac{2}{\pi}} \int_0^\infty f(t) \cos st \, dt$$

$$= \sqrt{\frac{2}{\pi}} \int_0^a \cos st \, dt$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{\sin st}{s} \right]_0^a$$

$$= \sqrt{\frac{2}{\pi}} \frac{\sin as}{s}$$

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3. $F_s\{f(x)\}$ $\quad f(x) = \dfrac{x}{x^2+1}$

$$F_s\{f(x)\} = \sqrt{\frac{2}{\pi}} \int_0^\infty \frac{x \sin x}{x^2+1}\, dx$$

$$I(s) = \sqrt{\frac{2}{\pi}} \int_0^\infty \frac{x \sin sx}{x^2+1}\, dx$$

$$I(s) = \sqrt{\frac{2}{\pi}} \int_0^\infty \frac{x^2 \sin sx}{x(x^2+1)}\, dx$$

$$= \sqrt{\frac{2}{\pi}} \left[\int_0^\infty \frac{(x^2+1)-1}{x(x^2+1)} \sin sx\, dx \right]$$

$$= \sqrt{\frac{2}{\pi}} \left[\int_0^\infty \frac{\sin sx\, dx}{x} - \frac{\sin sx}{x(x^2+1)}\, dx \right]$$

$$= \int_0^\infty \frac{\sin t}{t}\, dt = \frac{\pi}{2}$$

$t = sx$

$dt = s\,dx$

$dx = \dfrac{dt}{s}$

$$\int_0^\infty \frac{\sin sx}{sx}\, s\,dx = \pi/2$$

$$I(s) = \sqrt{\frac{2}{\pi}} \left[\frac{\pi}{2} - \int_0^\infty \frac{\sin sx}{x(x^2+1)}\, dx \right]$$

$$I'(s) = \sqrt{\frac{2}{\pi}} \left[\int_0^\infty -\frac{x \cos sx}{x(x^2+1)}\, dx \right]$$

$$= -\sqrt{\frac{2}{\pi}} \left[\int_0^\infty \frac{\cos sx}{(x^2+1)}\, dx \right]$$

$$= \sqrt{\frac{2}{\pi}} \left[\int_0^\infty \frac{x \sin sx}{x^2+1}\, dx \right]$$

$$I''(s) = I_s$$

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$\lambda^2 = 1 \qquad \lambda = \pm 1$

$I(s) = c_1 e^{\lambda_1 s} + c_2 e^{\lambda_2 s}$

$I(s) = c_1 e^{s} + c_2 e^{s} \qquad I(0) = c_1 + c_2$

$I'(s) = c_1 e^{s} - c_2 e^{-s} \qquad I'(0) = c_1 - c_2$

$I(0) = \sqrt{\frac{\pi}{2}}$

$I(0) = -\sqrt{\frac{2}{\pi}} \int_0^\infty \frac{1}{x^2+1}\, dx = -\sqrt{\frac{2}{\pi}} \left[\tan^{-1} x \right]_0^\infty$

$= -\sqrt{\frac{\pi}{2}}$

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28/Sep/2020

1. Find the fourier Trans transform.

$$f(x) = \begin{cases} 1 - |x| & \text{for } |x| < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Prove that $\int_0^\infty \frac{\sin^4 x}{x^4} dx$

Ans $1 - |x|$ is even so $F\{f(t)\} = F_c\{f(t)\}$

$$F_c\{f(t)\} = \sqrt{\frac{2}{\pi}} \int_0^\infty (1-x) \cos sx \, dx.$$

$\sqrt{\frac{2}{\pi}}$ / $1 < x$

$$= \sqrt{\frac{2}{\pi}} \left[(1-x) \frac{\sin sx}{s} - \frac{(-1) - \cos sx}{s^2} \right]_0^1$$

$$= \sqrt{\frac{2}{\pi}} \left[-\frac{\cos s}{s^2} - \left[\frac{\sin 0}{s} - \frac{\cos 0}{s^2} \right] \right]$$

$$\sqrt{\frac{2}{\pi}} \left[-\frac{\cos s}{s^2} + \frac{1}{s^2} \right]$$

$$F_s = \left(\frac{1 - \cos s}{s^2} \right) \sqrt{\frac{2}{\pi}}$$

$$\cos s = 1 - 2\sin^2 s/2$$

$$2\sin^2 s/2 = 1 - \cos s$$

$$= \sqrt{\frac{2}{\pi}} \frac{1}{2} \frac{\sin^2 s/2}{(s/2)^2}$$

$$= \frac{1}{\sqrt{2\pi}} \frac{\sin^2 s/2}{(s/2)^2}$$

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$$\int_0^1 (1-x)^2\,dx = \int_0^\infty (I_c(x))^2\,dx$$

$$\frac{1}{2\pi}\left(\frac{\sin^2 s/2}{(s/2)^2}\right)^2 ds$$

$s/2 = x$

$ds = 2dx$

$$\frac{2}{2\pi}\left(\frac{\sin^2 x}{x^2}\right)^2 dx$$

$$\frac{1}{\pi}\left(\frac{\sin^4 x}{x^4}\right) dx = \int_0^1 (1-x)^2\,dx \quad \nearrow 1/3$$

$$\frac{\sin^4 x}{x^4}\,dx = \frac{\pi}{3}$$

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Find the Inverse Fourier Transform of

$$F(s) = \begin{cases} a - |s| & \text{for } |s| < a \\ 0 & \text{for } |s| > a \end{cases}$$

and hence evaluate $\int_0^\infty \left(\frac{\sin x}{x}\right)^2 dx$.

$$F^{-1}[F(s)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ist} F(s)\, ds.$$

$$= \frac{1}{\sqrt{2\pi}} \left[\int_{-a}^{a} e^{ist} \, a - |s| \, ds \right]$$

$$= \frac{1}{\sqrt{2\pi}} \left[\int_{-a}^{a} (\cos st + i \sin st) \, a - |s| \, ds. \right]$$

$$= \frac{1 \times 2}{\sqrt{2\pi}} \left[\int_{0}^{a} \cos st \, a - |s| \, ds \right]$$

$$= \frac{2}{\sqrt{2\pi}} \left[\int_{-a}^{a} \cos st \, a \, ds - \int_{-a}^{a} s \cos st \, ds \right]$$

$$= \frac{1}{\sqrt{2\pi}} 2 \left[\frac{a \sin st}{t} \right]_0^a - \int_{-a}^{a} \left[\frac{s \sin st}{t} + \frac{\cos st}{t^2} \right]_0^a$$

$$= \frac{1}{\sqrt{2\pi}} 2 \left[\frac{a \sin at}{t} + \frac{a \sin 0}{t} - \right.$$

$$\left. \left[\frac{a \sin at}{t} + \frac{\cos at}{t^2} - \frac{1}{t^2} \right] \right]$$

$$= \frac{1}{\sqrt{2\pi}} 2 \left[\cancel{\frac{a \sin at}{t}} - \cancel{\frac{a \sin at}{t}} - \frac{\cos at}{t^2} + \frac{1}{t^2} \right]$$

$$= \frac{1}{\sqrt{2\pi}} 2 \left[\frac{1}{t^2} - \frac{\cos at}{t^2} \right]$$

classmate
Date
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Cos

$$f(t) = \frac{\sqrt{2}}{\sqrt{\pi}\, t^2} - \left[\cos at - 1\right]$$

$$F\{f(t)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ist} f(t)\, dt$$

$$= \frac{1}{\sqrt{2}\pi} \int_{-\infty}^{\infty} e^{-ist} \cancel{f(t)/dt} \frac{\sqrt{2}}{\sqrt{\pi}\, t^2} \cos at - 1$$

$$= \frac{1}{\pi} \int_{-\infty}^{\infty} e^{-ist} \frac{-(\cos at - 1)}{+ t^2} \qquad \left.\frac{\cos at - 1}{t^2}\right\} \rightarrow \text{Even}$$

$$= \frac{1}{\pi}\, 2 \int_{0}^{\infty} \cos st \frac{-(\cos at - 1)}{t^2}\, dt$$

$$= \frac{1}{\pi}\, 2 \int^{\infty} \cos st \frac{-(\cos at)}{t^2}\, dt \qquad \int^{\infty} \frac{\cos at - 1}{t^2}\, dt$$

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$$F(s) = \frac{2}{\pi}\int_0^\infty \cos st \; 2 \frac{\sin^2 at}{t^2}\, dt$$

$$s = 0 \quad a = 2$$

$$2 = \frac{4}{\pi}\int_0^\infty \frac{\sin^2 t}{t^2}\, dt$$

Susanth
2/4/2020

$$\frac{\pi}{2} = \int_0^\infty \frac{\sin^2 t}{t^2}\, dt.$$

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